

Upper Minkowski dimension need not lower-bound the power-exponential size of a Wasserstein space

Candidate counterexample to the strongest conjecture in arXiv:1304.5219

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Status: candidate full counterexample, likely valid. For every prescribed finite $D > 0$ there is a compact ultrametric space X such that

$$\overline{\dim}_M X = D \quad \text{but} \quad \text{crit}_{\mathcal{P}}(\mathcal{P}(X), W_p) = 0 \quad \text{for every } p \in [1, \infty).$$

Thus even among compact ultrametric spaces, upper Minkowski dimension does not in general lower-bound the power-exponential critical parameter of the Wasserstein space. In particular, the strongest conjecture in the source paper is false.

1 The source question

Kloeckner proves for every compact metric space X that

$$\dim_H X \leq \text{crit}_{\mathcal{P}}(\mathcal{P}(X), W_p) \leq \overline{\dim}_M X.$$

At the end of the paper he asks whether, at least for ultrametric X , the critical parameter might also be bounded below by the lower or upper Minkowski dimension, and records as the strongest conjecture

$$\text{crit}_{\mathcal{P}}(\mathcal{P}(X), W_p) = \overline{\dim}_M X \quad \text{for every compact metric } X \quad (\text{Q})$$

[1, p. 21, final paragraph]. Here is the source passage.

In conclusion, X is a countable ultrametric space whose Wasserstein space has infinite power-exponential critical parameter, as claimed.

It seems plausible that (maybe at least for ultrametric spaces) the critical parameter of $\mathcal{W}_p(X)$ is bounded below by the (lower or upper?) Minkowski dimension of X . The strongest conjecture would be that $\text{crit}_{\mathcal{P}} \mathcal{W}_p(X) = \overline{\text{M}}\text{-dim } X$ for all compact metric space X . However we do not know how to transform an ultrametric space into one where explicit computations are possible without losing too much of its Minkowski dimension, and general ultrametric spaces seem difficult to handle in wide generality. Moreover, we do not know whether there is an Ultrametric Skeleton Theorem with respect to Minkowski dimension (see Question 1.11 in [MN13b]).

References

We disprove the upper-Minkowski lower bound, already in the ultrametric class. Our example has lower Minkowski dimension zero, so it does not decide the weaker possible lower-Minkowski bound.

2 Definitions and a covering criterion

For a compact metric space Y , let $N_Y(r)$ denote the least number of radius- r balls that cover Y . We use

$$\overline{\text{dim}}_M Y = \limsup_{r \downarrow 0} \frac{\log N_Y(r)}{\log(1/r)}.$$

For $s > 0$, put $\psi_s(r) = \exp(-r^{-s})$. The power-exponential critical parameter is the critical value of the Hausdorff scale $(\psi_s)_{s>0}$. We need only the following direct consequence of the definition.

Lemma 1 (Cheap subsequential covers). *Suppose Y has, for a sequence $r_k \downarrow 0$, a cover by Q_k balls of radius r_k such that*

$$\log Q_k - (2r_k)^{-s} \longrightarrow -\infty.$$

Then the ψ_s -Hausdorff measure of Y is zero. If this holds for every $s > 0$, then $\text{crit}_{\mathcal{P}} Y = 0$.

Proof. The covering balls have diameter at most $2r_k$, and the sum of their ψ_s -costs is at most

$$Q_k \psi_s(2r_k) = \exp(\log Q_k - (2r_k)^{-s}) \longrightarrow 0.$$

Since their diameters tend to zero, the Hausdorff outer measure is zero. The final assertion follows by letting $s > 0$ be arbitrary. $\square$

We also use an elementary finite-simplex estimate.

Lemma 2 (A net for finitely supported measures). *Let Z be an n -point metric space of diameter at most 1. For every $p \geq 1$ and $0 < r < 1$, $(\mathcal{P}(Z), W_p)$ has an $r/2$ -net of cardinality at most*

$$(1 + 2^p n r^{-p})^n. \tag{1}$$

Proof. Enumerate $Z = \{z_1, \dots, z_n\}$ and set $\delta = (r/2)^p/n$. Given a probability vector (a_i) , round each of $a_1, \dots, a_{n-1}$ down to a multiple of δ and put the remaining mass at z_n . The original and rounded vectors have total variation distance at most $n\delta = (r/2)^p$. A maximal coupling moves only this amount of mass, by distance at most 1, so its p -cost is at most $(r/2)^p$. There are at most $(1 + \delta^{-1})^{n-1}$ rounded vectors, which is bounded by (1). $\square$

3 Proof intuition

Upper Minkowski dimension is a limsup. We exploit this by placing a huge finite equilateral block at a radius R_k , large enough that the covering number near R_k looks D -dimensional, and then leaving a vastly larger gap before the next radius R_{k+1} . These isolated bursts force $\overline{\dim}_M X = D$.

The Wasserstein space still contains a large finite simplex at every burst, but a finite simplex becomes inexpensive to cover if one is allowed to go far enough below its diameter. At a radius chosen deep in the gap (R_{k+1}, R_k) , all later mass can be collapsed to the accumulation point at negligible transport cost, while the first k blocks form only one finite simplex. Its logarithmic covering cost grows merely as its dimension times the logarithm of the requested accuracy. The power-exponential gauge at the same accuracy is double-exponentially smaller. A sequence of such cheap covers makes every positive power-exponential Hausdorff measure vanish.

4 The ultrametric counterexample

Theorem 3. *For every $D \in (0, \infty)$ there is a compact countable ultrametric space X_D satisfying*

$$\overline{\dim}_M X_D = D, \quad \text{crit}_{\mathcal{D}}(\mathcal{P}(X_D), W_p) = 0 \quad \text{for every } p \in [1, \infty).$$

The same space X_D works simultaneously for all p .

Proof. Fix $D > 0$. Define

$$A_1 = 1, \quad A_{k+1} = \exp(4(D+1)A_k), \quad R_k = e^{-A_k}, \quad M_k = \lceil e^{DA_k} \rceil. \quad (2)$$

Let C_k be a set of M_k points, take all the C_k disjoint, and set

$$X_D = \{\infty\} \sqcup \bigsqcup_{k \geq 1} C_k.$$

Give a point of C_k radius R_k and define, for distinct finite points,

$$d(x, \infty) = R_k \quad (x \in C_k), \quad d(x, y) = \max\{d(x, \infty), d(y, \infty)\}. \quad (3)$$

For any three distinct points the two largest pairwise distances coincide, so d is an ultrametric. Since $R_k \downarrow 0$, every tail $\{\infty\} \cup \bigcup_{i > k} C_i$ lies in the radius- R_{k+1} ball about ∞ ; hence X_D is compact.

Put $S_k = \sum_{i \leq k} M_i$. Super-exponential growth in (2) gives

$$\log(S_k + 1) = DA_k + o(A_k). \quad (4)$$

If $R_{k+1} \leq r < R_k$, each point in $C_1 \cup \dots \cup C_k$ needs its own r -ball, while the remaining tail needs one, and therefore

$$N_{X_D}(r) = S_k + 1. \quad (5)$$

Equations (4)–(5) give the upper bound $\overline{\dim}_M X_D \leq D$. Taking $r = R_k/2$ gives the reverse inequality, since $R_{k+1} < R_k/2$ for all large k . Thus

$$\overline{\dim}_M X_D = D. \quad (6)$$

It remains to estimate the Wasserstein space. Fix $p \geq 1$; the construction of X_D does not depend on this choice. Let

$$Z_k = \{\infty\} \cup \bigcup_{i \leq k} C_i, \quad n_k = |Z_k| = S_k + 1, \quad \rho_k = \exp(-\sqrt{A_{k+1}}). \quad (7)$$

Collapse $X_D \setminus Z_k$ to ∞ and denote this retraction by π_k . Coupling every point with its image shows, for every $\mu \in \mathcal{P}(X_D)$,

$$W_p(\mu, (\pi_k)_\# \mu)^p \leq \int d(x, \pi_k x)^p d\mu(x) \leq R_{k+1}^p. \quad (8)$$

For large k , $R_{k+1} \leq \rho_k/2$. Since $\text{diam } X_D < 1$, Lemma 2 supplies a $\rho_k/2$ -net of $\mathcal{P}(Z_k)$ with at most

$$Q_k := (1 + 2^p n_k \rho_k^{-p})^{n_k} \quad (9)$$

points. Combining this net with (8) shows that the same centers form a ρ_k -net of $\mathcal{P}(X_D)$ in W_p .

Now (4) and (7) imply

$$\begin{aligned} \log Q_k &\leq n_k (O_p(1) + \log n_k + p \log(1/\rho_k)) \\ &= O_p(e^{DA_k} \sqrt{A_{k+1}}) = O_p(e^{(3D+2)A_k}), \end{aligned} \quad (10)$$

because $\sqrt{A_{k+1}} = e^{2(D+1)A_k}$. On the other hand, for every fixed $s > 0$,

$$(2\rho_k)^{-s} = 2^{-s} \exp(s\sqrt{A_{k+1}}) = 2^{-s} \exp(se^{2(D+1)A_k}). \quad (11)$$

The quantity in (11) dominates (10), so

$$\log Q_k - (2\rho_k)^{-s} \longrightarrow -\infty.$$

Lemma 1 gives zero ψ_s -Hausdorff measure for $(\mathcal{P}(X_D), W_p)$. This holds for every $s > 0$, hence its power-exponential critical parameter is zero. Since $p \geq 1$ was arbitrary, the conclusion holds for all p on the same X_D . $\square$

5 Consequences, checks, and scope

Taking any $D > 0$ in Theorem 3 yields

$$\text{crit}_{\mathcal{D}}(\mathcal{P}(X_D), W_p) = 0 < D = \overline{\dim}_M X_D,$$

which directly refutes the upper-Minkowski lower bound for compact ultrametrics and therefore refutes (Q).

The proof uses no unproved regularity or selection lemma. The two estimates that carry the argument are independently auditable: tail collapse costs at most R_{k+1} by the explicit coupling (8), and rounding a finite probability vector gives the net (1). Constants such as 2 are immaterial but have been retained so that the Hausdorff-gauge cover is literal rather than asymptotic.

The example deliberately has

$$\underline{\dim}_M X_D = 0.$$

Indeed, at the intermediate radii $r_k = \exp(-\sqrt{A_k A_{k+1}})$, equation (5) gives

$$\frac{\log N_{X_D}(r_k)}{\log(1/r_k)} = \frac{DA_k + o(A_k)}{\sqrt{A_k A_{k+1}}} \longrightarrow 0.$$

Thus this packet does not disprove a lower bound by *lower* Minkowski dimension, nor does it address an ultrametric skeleton theorem formulated with that invariant.

Novelty check. Before promotion, bounded searches were run through 11 August 2026 for the exact title and conjectural formula, for combinations of “Wasserstein”, “power-exponential critical parameter”, “upper Minkowski dimension”, and “ultrametric”, and for later work citing the source question. No paper explicitly answering this conjecture was located. The July 2026 arXiv paper *Regular ultrametric skeletons* was also checked at the abstract/search level; its stated result concerns measure-controlled regular skeletons and did not reveal this Wasserstein counterexample. Novelty confidence is moderate because the search was bounded rather than exhaustive.

Human-review recommendation. Check the convention relating covering balls to Hausdorff diameters in Lemma 1, the exact covering count (5), the rounding and maximal-coupling bound in Lemma 2, and the separation of single- versus double-exponential growth in (10)–(11).

References

- [1] B. R. Kloeckner, *A geometric study of Wasserstein spaces: ultrametrics*, Mathematika 61 (2015), no. 1, 162–178; arXiv:1304.5219, <https://arxiv.org/abs/1304.5219>.